Task 1: Quality Check Bug Report & Root Cause Analysis

Target App: RealWorld Conduit (demo.realworld.show)

Tester	prasanna	**Test date**	10 November 2026
Article slug	mastering-web-automation-and-qa-testing	**Browser**	Google Chrome (desktop)

1. Core Flow Execution Summary

A Quality Check audit was carried out on https://demo.realworld.show covering the six core flows. All six flows completed successfully with valid input. The test article **"Mastering Web Automation and QA Testing"** was published under author **prasanna** with the tags **qa, automation, testing** (after fixing the tag-entry problem in BUG-02), then edited so the tags became **automation, quality, testing** (the tag *qa* was replaced by *quality*), and finally deleted. Edge cases found during the audit are logged in Section 2.

Test Flow	Execution Details & Validation Path	Status
1. Sign-Up	Registered a new user with valid credentials; account created successfully.	**PASS**
2. Login / Authentication	Signed in with the registered account (prasanna); session established and the Global Feed loaded with the user shown in the header.	**PASS**
3. Create Content	Published the article with title, description, markdown body and tags. The first attempt failed on tag validation (BUG-02); it succeeded once each tag was added with **Enter**. Markdown rendered correctly (headings, bold list items).	**PASS** (after BUG-02 workaround)
4. Edit Content	Edited the published article; the tag *qa* was changed to *quality* and the change persisted on the article page.	**PASS**
5. Delete Content	Deleted the article; the record was removed and the user was redirected cleanly.	**PASS**
6. Logout	Clicked Logout; the session was cleared and protected routes were no longer accessible.	**PASS**

2. Quality Check Bug Report & Observations

Bug ID	Title	Flow	Steps to Reproduce	Actual Behavior	Severity
BUG-01	Missing client-side email format validation	Sign-Up	1. Open the Sign Up form. 2. Enter a malformed email (user@domain). 3. Click Submit.	The request is sent to the backend with no inline highlighting on the field; the backend returns a 422 error.	**MEDIUM**
BUG-02	Tag field does not accept comma-separated input; raw backend error shown	Create Content	1. Open New Article and fill title, description and body. 2. In the tags field type `qa, automation, testing`. 3. Click Publish Article.	The whole text is stored as **one** tag (23 characters), which breaks the rule of fewer than 20 characters per tag. Publish fails and an unformatted message appears at the top of the form: *"body tagList is an optional list of less than 10 strings of less than 20 chars"*. The field gives no hint that Enter is needed. (Evidence: Screenshot 1)	**MEDIUM**
BUG-03	Stale authenticated state after logout	Logout	1. Log in as prasanna. 2. Click Logout. 3. Press the browser Back button.	The cached feed and the authenticated header (user name, New Article, Settings) briefly flash before the app redirects.	**MEDIUM**

Bug ID	Title	Flow	Steps to Reproduce	Actual Behavior	Severity
BUG-04	Blank / whitespace-only tag accepted	Create Content	1. Enter space-only characters in the tag field. 2. Click Publish Article.	Empty tag badges are rendered next to the valid tags.	**LOW**
BUG-05	Unhandled 404 on direct navigation to a deleted article	Navigation	1. Delete the article. 2. Open the direct URL /article/mastering-web-automation-and-qa-testing.	The page stays on a permanent loading state and a JavaScript runtime error occurs instead of a "not found" page.	**MEDIUM**
BUG-06	Markdown blockquote not styled	Create Content	1. In the article body type `> "Quality is not an act, it is a habit."` 2. Publish and open the article.	The line is shown as plain quoted text; there is no blockquote styling (no indent or left border), unlike the headings and bold list items which render correctly. (Evidence: Screenshot 2)	**LOW**

Severity note: BUG-05 is rated MEDIUM because it only affects deleted or invalid article URLs. It should be raised to HIGH if the loading hang is shown to freeze the whole application.

Positive observations

- The form keeps the entered title, description and body after the tag error, so no work is lost.
- Edit and Delete controls are shown only to the article owner.
- Edits to tags and content persist correctly after saving.

3. Root Cause Analysis

3.1 BUG-02: Tag input delimiter

Observation: Entering `qa, automation, testing` in the tags field and publishing fails with a validation message shown as raw text at the top of the form.

Root cause: The tags field is a tag-chip input where each tag is committed with the **Enter** key. Commas are not treated as separators, so the full string became a single tag of 23 characters. The API rule allows at most 9 tags (fewer than 10), each shorter than 20 characters, so the request was rejected. The front end passes the backend message to the page without formatting it.

Recommended fix: (a) Split the input on commas when a comma is typed or pasted. (b) Add placeholder text such as "Press Enter to add a tag". (c) Validate tag length on the client and show a readable message beside the field.

3.2 BUG-05: Deleted article URL

Observation: Opening the URL of a deleted or non-existent article leaves the page in a loading state.

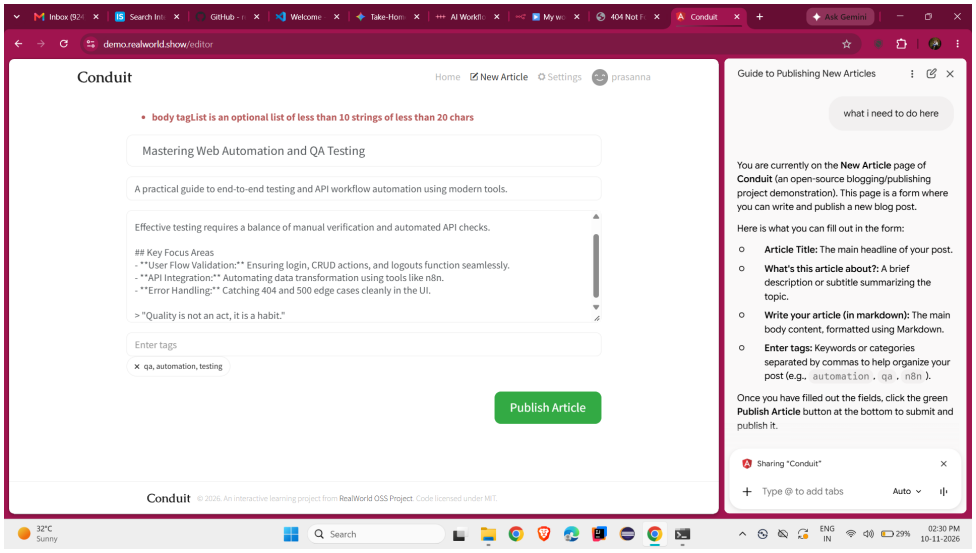
Likely root cause (inferred from the behavior; to be confirmed with the browser console error): the article loader reads the result directly (for example `response.data.article`). When the backend returns HTTP 404 for a missing slug, the rejection handler does not redirect or reset the loading flag, and a `TypeError: Cannot read properties of undefined` is probably thrown.

Recommended fix: Add an explicit status check in the article loader and always clear the loading state:

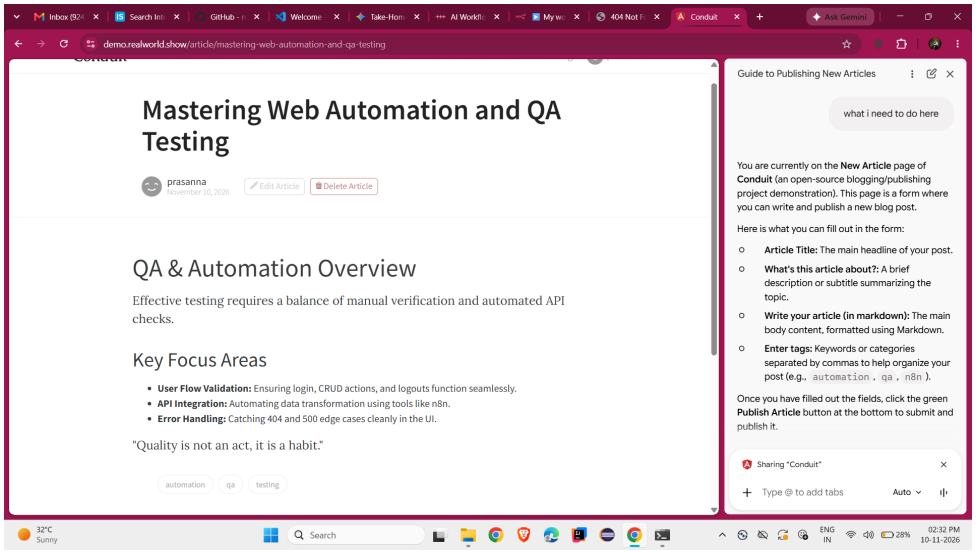
```
api.getArticle(slug)
  .then(article => setArticleState(article))
  .catch(err => {
    if (err.response?.status === 404) {
      navigate('/404');
    }
  })
  .finally(() => setLoading(false));
```

4. Evidence

Screenshot 1: BUG-02. Publish fails with the raw validation message; the typed text became one tag chip.



Screenshot 2: Published article. The headings and list render correctly, but the blockquote line is plain text (BUG-06).



Screenshot 3: After editing. The tag *qa* was changed to *quality* (tags: automation, quality, testing).

